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packet :  $\Pi = (\text{page}, \text{type}, \varphi_0, \text{intent}, \text{sources}, \text{constraints}, \text{gate}, \text{limits})$ 
receipts:  $\rho = (r_1, \dots, r_n)$ , one receipt per attempted phase
1  $\varphi \leftarrow \varphi_0$  // an existing Page enters at CHECK, a new one at
   OUTLINE
2  $s \leftarrow 0$ ;  $k \leftarrow 1$  // steps taken, and the current round
3 while  $s < \text{limits.max\_steps}$  and  $k \leq \text{limits.max\_rounds}$  do
4    $\Gamma \leftarrow \text{CONTEXT}(\text{page}, \varphi)$  // one hop, this phase's rows only;
   a bad row is a HOLD
5   if  $\varphi \neq \text{CHECK}$  then
6      $(v', \hat{r}) \leftarrow \text{Producer}(\varphi, \Pi, \Gamma)$  // one phase, by one actor,
     who also proposes a route
7      $v' \leftarrow \text{Builder}(v')$  // rebuild, deterministic checks,
     version identity
8   else
9      $(v', \hat{r}) \leftarrow \text{Judge}(v, \Pi, \Gamma)$  // a fresh read-only actor,
     judging one exact version
10   $r \leftarrow \text{RECEIPT}(\varphi, \text{actor}, v \rightarrow v', \hat{r})$ ; append  $r$  to  $\rho$ 
11  if  $\hat{r}.\text{route} \notin R(\varphi)$  then return  $\rho$  with HOLD
12  if  $\hat{r}.\text{route} \in \{\text{CLOSE}, \text{HOLD}\}$  then return  $\rho$ 
13  if  $\hat{r}.\text{route} = \text{DRAFT}$  and  $\varphi \notin \{\text{OUTLINE}, \text{DRAFT}\}$  and  $\hat{r}.\text{reopens}$ 
    then  $k \leftarrow k + 1$ 
14   $\varphi \leftarrow \hat{r}.\text{route}$ ;  $v \leftarrow v'$ ;  $s \leftarrow s + 1$ 
15 return  $\rho$  with LIMITREACHED // did not converge, which never
    means the Page passed

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